

Policy Memo

To: Minister of Investment, Trade and Industry of Malaysia

From: Aldiyar Auyezbek, Policy Adviser

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Subject: Move Malaysia-Singapore semiconductor cooperation from production to joint R&D

Summary

Malaysia and Singapore already share parts of one semiconductor production chain. STMicroelectronics, for example, fabricates wafers in Singapore and sends them to Johor for assembly, packaging and qualification. Singapore [describes](#) the JS-SEZ model in similar terms: firms can keep headquarters, innovation, R&D and IP in Singapore while using Johor to expand production.

This arrangement brings investment into Malaysia. It can also create an awkward division of labour. Malaysia's National Semiconductor Strategy (NSS) [wants](#) to [move](#) beyond manufacturing into technology and R&D, including Malaysian firms in IC design, advanced packaging and advanced equipment. Singapore, meanwhile, openly wants to keep R&D, IP and other high-value functions at home.

Malaysia should propose a **several-year joint semiconductor R&D programme** with Singapore. Each government would fund firms and research teams from its own country, but selected projects would use researchers, companies and facilities on both sides. Malaysian funding should only go to projects that leave research or engineering capability in Malaysia.

Background

The starting machinery is already there. The JS-SEZ has a Malaysian investment facilitation centre, a Singapore project office and ministerial oversight. Malaysia and Singapore [have agreed](#) to attract 50 projects in five years and 100 projects in ten. Singapore and Singapore-based firms had committed more than S\$5.5 billion to Johor by July 2026.

Semiconductors are already part of this cooperation. In May 2026, MIDA used a Micron-OCBC JS-SEZ supplier event to connect Malaysian firms with semiconductor supply-chain opportunities. This is [useful](#) supplier development, but it is still mainly about firms entering existing supply chains.

There is also a bilateral business fund. The Malaysia-Singapore Business Development Fund [pays](#) up to 50% of eligible costs for joint feasibility studies, missions and pilots, capped at **RM200,000 per activity**. MITI [said](#) in October 2025 that it wanted to refocus the fund on JS-SEZ priorities, including joint ventures and technology exchange.

The published programmes still stop short of jointly developing semiconductor technology. This is striking because both governments are now spending on some of the same broad technology areas. Malaysia's Advanced Packaging Consortium [has](#) **RM92 million in public R&D funding matched by RM93.8 million from industry**. Singapore has [committed](#) **S\$800 million** to its semiconductor RIE Flagship and **S\$500 million** to a new National Semiconductor Translation and Innovation Centre R&D fab, which starts operations in 2027 with shared 12-inch research tools and an initial focus on advanced packaging.

Analysis and options

Option 1: keep the current twinning model

There is a reasonable case for leaving the division of work to firms. Singapore [has](#) strong research institutions, advanced fabs, finance and multinational headquarters. Malaysia has manufacturing scale, packaging and testing operations and a large supplier base. Firms such as STMicroelectronics already combine these assets without a joint government R&D programme.

The problem is that Singapore [has](#) been quite clear about what it wants from this division. In July, its Ministry of Trade and Industry said Singapore should keep HQ functions, R&D, IP and other high-value activities while firms use Johor to scale. On 10 September, MTI [said](#) Singapore grants could support R&D undertaken in the JS-SEZ, but only where there was a clear route for capabilities and economic value to return to Singapore.

Malaysia [wants](#) some of those same activities. In February 2025, MITI approved plans to develop 110 Malaysian semiconductor companies in IC design, advanced packaging and advanced manufacturing equipment. The same policy states that Malaysia wants to move from a manufacturing-based model towards technology, R&D and “technological sovereignty”.

If the current division becomes permanent, Malaysia could get more semiconductor investment without getting much closer to that goal.

Option 2: add joint R&D to production twinning

Malaysia could ask Singapore to run one semiconductor technology programme jointly. Start with one field where firms on both sides already have something to contribute. Advanced packaging, testing or semiconductor equipment are obvious candidates, although the first technology should be chosen with industry rather than written into the agreement now.

The case for government involvement comes from the cost of research infrastructure and the gap between research and production. Singapore is [spending](#) S\$500 million on a shared R&D fab precisely because individual start-ups and SMEs cannot easily afford industry-grade 12-inch tools. Malaysia already [has](#) firms working on packaging, test and equipment that could participate in taking prototypes towards industrial use.

Malaysia therefore faces an odd choice. It can spend heavily to recreate research infrastructure already available across the Causeway, or it can let Malaysian firms enter only after Singaporean research has moved closer to production. Joint projects offer a third route: use some infrastructure together while Malaysian engineers participate before the technology reaches the factory.

Cross-border industrial policy already exists elsewhere. The EU's second microelectronics IPCEI [links](#) **68 R&D and first-industrial-deployment projects across 14 member states**. Governments finance participating firms under one wider technology programme; the scheme permits up to €8.1 billion in public support and expects €13.7 billion in private investment. Malaysia and Singapore need nothing of this size. The useful part is the structure: national financing can support a shared technology programme.

Malaysia could still lose from such an arrangement. Access to Singapore's laboratories does not guarantee that Malaysian firms acquire useful technology. Malaysian engineers could move across the border. Governments could also end up paying for collaboration that firms would have undertaken themselves. These are reasons to make Malaysian capability a condition of funding, rather than to subsidise cross-border activity in general.

Recommendation

MITI should propose a **three-year Malaysia-Singapore Semiconductor R&D Programme** through the existing bilateral ministerial machinery. It should start with one technology programme, not a new general semiconductor fund.

1. **Each government pays for its own side.** Malaysia finances Malaysian firms and research teams; Singapore does the same for its participants. Every project needs private co-funding and firms or research teams from both countries.
2. **Share expensive facilities where it makes sense.** Malaysian project teams should be able to use selected Singapore public semiconductor R&D infrastructure. Singapore participants should have access to relevant Malaysian packaging,

test and equipment facilities. Participation from Malaysia should include Penang and Kulim, where much of its semiconductor base sits, instead of restricting the programme to Johor.

3. **Malaysia pays for capability, not participation.** Before receiving Malaysian funding, a project must specify what will remain in Malaysia after the grant ends: for example, a Malaysian engineering team, jointly developed IP, a qualified production process, a new product or a local supplier able to sell the resulting technology. A project that leaves Malaysia only with assembly or production work should not qualify.

The JS-SEZ has already shown that Malaysia and Singapore can organise production across a border. The harder experiment is to see whether the two governments can do the same with semiconductor research, while giving both countries a reason to stay in the arrangement.